

STEEL STRUCTURE CONVERSION

E3D MAC → Tekla IFC

Converts an E3D MAC into Tekla IFC (reverse of TEKLATOE3D). SCTN/PANE → IfcBeam/Plate.

Key Purpose

Why this program

1. Bring E3D steel back into Tekla — parametrically

– SCTN members become IfcBeam/Column/Member with real IfcIShape/UShape/LShape profiles, not dumb meshes

→ **Effect:** Tekla recognizes and can re-select each section size, so the imported frame stays fully editable instead of frozen geometry

2. One-file offline conversion, no extra software

– Pure standard-library Python reads the .MAC and writes IFC2X3 — no plugins, no libraries, no license

→ **Effect:** Any PC with base Python runs it, so model hand-off is instant and portable across machines

3. Carry more than just beams

– Panels become IfcPlate, foundation pads become IfcFooting, gratings become plates, and BOX/CYL/TORUS/DISH primitives ride along as solids

→ **Effect:** The whole structural assembly transfers in one pass, cutting manual re-modelling and transcription error

Required inputs : An E3D/PDMS export MAC file (NEW SCTN / BOX / CYLINDER / CTORUS / DISH with POS/ORI/SPRE)

Main Window – Item Guide

The screenshot shows the main window of the 'E3D MAC → Tekla IFC Converter' application. The window has a title bar with the application name and standard OS window controls. Below the title bar is an orange header bar with a gear icon and the text 'E3D MAC → Tekla IFC Converter'. The main content area is divided into several sections. At the top, there is a section titled 'Select E3D MAC File' (callout 1) containing a text input field (callout 2) and a 'Browse...' button (callout 3). Below this is an 'Options' section (callout 4) with a label 'Primitives (BOX/CYL/TORUS/DISH):' (callout 5) followed by a dropdown menu (callout 6) and a note '(Off = SCTN parametric members only)' (callout 7). The next section is 'Progress / Result' (callout 8), which contains a large empty rectangular area (callout 9). Below that is an 'Output Files' section (callout 10) with a list box (callout 11). At the bottom right, there is a large blue 'Convert' button (callout 12).

1. Select E3D MAC File
2. MAC file path input
3. Browse... for MAC file
4. Option section
5. Primitives (BOX/CYL/TORUS/DISH) On/Off
6. Primitives On/Off selector
7. Off = SCTN parametric members only
8. Progress / Result log
9. Conversion log output
10. Output Files list (double-click opens folder)
11. Output IFC file list
12. Convert

Overview

What it does

1. Converts an E3D MAC into Tekla IFC (reverse of TEKLATOE3D)
2. SCTN/PANE → IfcBeam/Plate

Category: Steel Structure Conversion

Folder: STEEL_STRUCTURE/E3DTOTEKLA

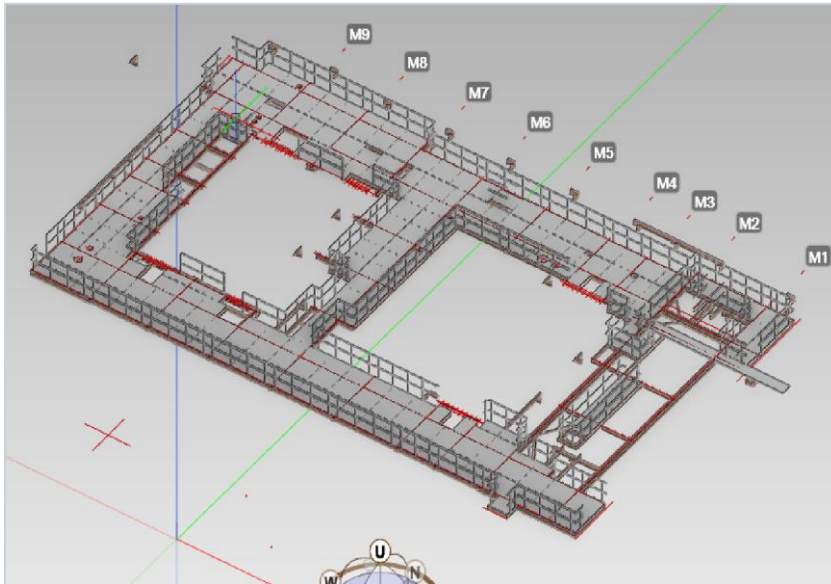
Main scripts: e3dtotekla.py

Scripts / Status: 2 scripts · Operational

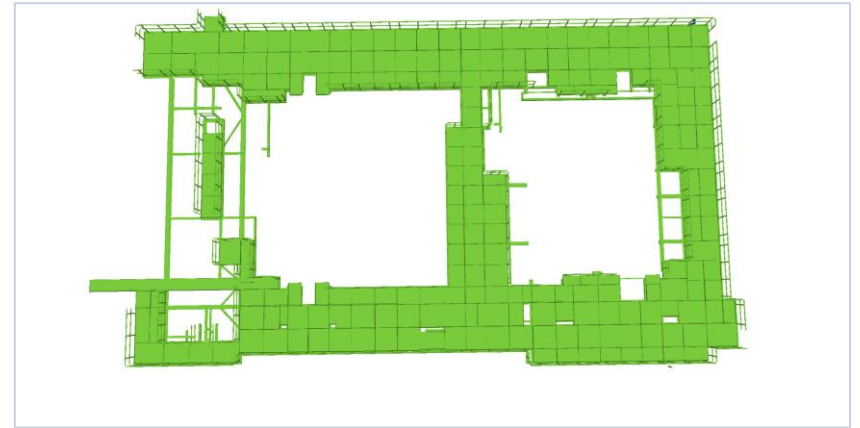
Tags: E3D · IFC · Tekla · SCTN

Output Samples

Output Samples (1/2)



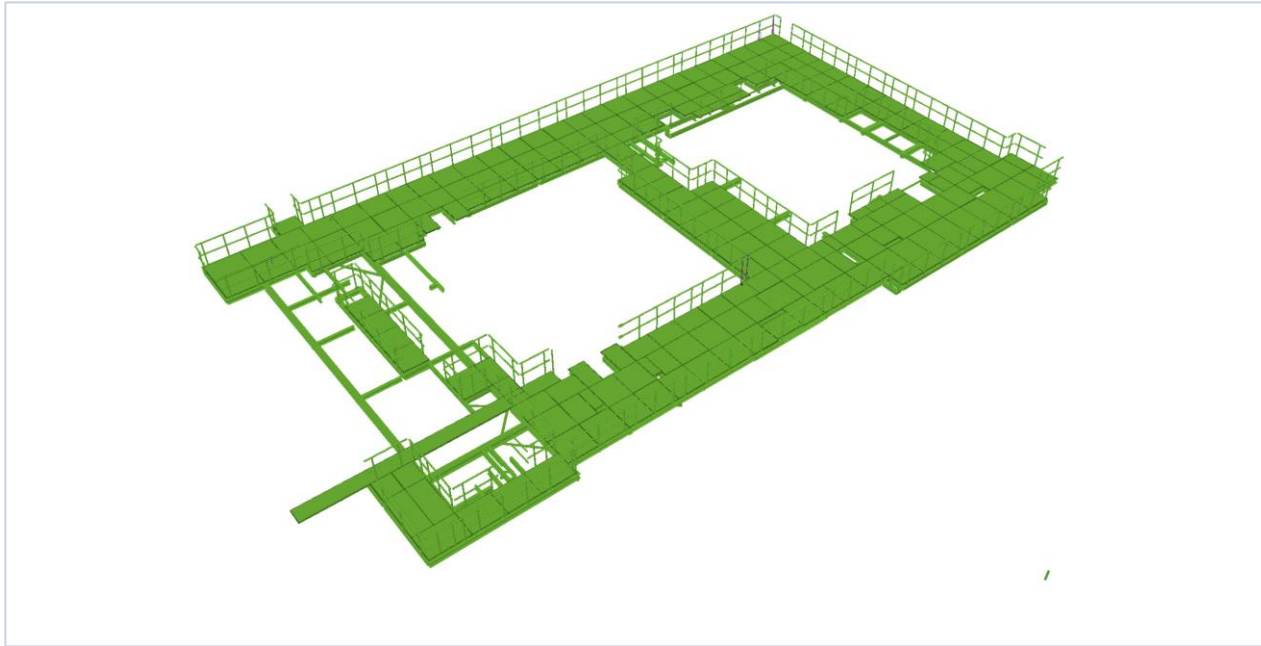
E3D to Tekla - E3D Input



E3D to Tekla - Tekla Output 1

Output Samples

Output Samples (2/2)



E3D to Tekla - Tekla Output 2

Notes & Information

- Part of the in-house Plant Engineering automation suite — category: Steel Structure Conversion.
- Runs offline as a pure-Python / Tkinter tool; portable across PCs via OneDrive sync (needs Python + packages).
- Launch from the PC launcher (런처.bat) or run the main script: e3dtotekla.py.
- Output samples and the program window above reflect the current build.

Thank You

E3D MAC → Tekla IFC · Plant Engineering Total Service System